

Pure O -sequences, introduced in the context of Stanley's work on Hilbert functions of monomial level algebras in the late 1970s, have been studied intensively over the last fifteen years [2]. They exhibit surprisingly strong positivity phenomena, especially in small codimension and type. Although pure O -sequences are not log-concave in general, the case of codimension 3 and type 2 remained open for over a decade [1, 3]. The following theorem establishes log-concavity in precisely this remaining case.

Terminology A monomial $x_0^{m(0)}x_1^{m(1)}x_2^{m(2)}$ in three variables is identified with the exponent vector $m = (m(0), m(1), m(2)) \in \mathbb{N}^3$. We write $m' \leq m$ for coordinate-wise inequality and $|m| = m(0) + m(1) + m(2)$ for total degree. A finite order ideal $\Gamma \subset \mathbb{N}^3$ is a finite set closed downward under this order: if $n \in \Gamma$ and $m \leq n$, then $m \in \Gamma$.

A monomial $m \in \Gamma$ is maximal if every $n \in \Gamma$ with $m \leq n$ also satisfies $n \leq m$. The ideal is pure if all maximal monomials have the same total degree; this common degree is the socle degree. The type is the number of maximal monomials. The pure O -sequence of Γ is

$$h_\Gamma(d) = \#\{m \in \Gamma : |m| = d\}.$$

It is log-concave if

$$h_\Gamma(i-1)h_\Gamma(i+1) \leq h_\Gamma(i)^2 \quad (0 < i < e).$$

where e is the socle degree of Γ .

Theorem (Hilbert functions) Every pure O -sequence of codimension 3 and type 2 is log-concave.

Proof. Let Γ be a pure order ideal in three variables and suppose that its two maximal monomials are $m_1 \neq m_2$. Since Γ is pure, $|m_1| = |m_2| = e$. Every element of a finite order ideal lies below a maximal element, so

$$\Gamma = \text{Div}(m_1) \cup \text{Div}(m_2), \text{ where } \text{Div}(g) = \{u : u \leq g\}.$$

After possibly interchanging m_1 and m_2 , there is a coordinate $x \in \{0, 1, 2\}$ such that

$$m_2(x) > m_1(x), \quad m_2(y) \leq m_1(y) \quad (y \neq x). \quad (1)$$

Indeed, the three integers $m_1(i) - m_2(i)$ sum to zero and are not all zero, hence one sign occurs in a minority position; if necessary we swap the two monomials so that the minority sign is negative. Put

$$c = m_1(x) + 1, \quad M_{red}(y) = \begin{cases} m_2(x) - m_1(x) - 1, & y = x, \\ m_2(y), & y \neq x. \end{cases}$$

Then $|M_{red}| = e - c$. For g a monomial, write

$$h_g(d) = \#\{u \leq g : |u| = d\}.$$

For $u \leq m_2$, condition (1) implies

$$u \not\leq m_1 \iff u(x) \geq c.$$

If $d \geq c$, the map $u \mapsto u - ce_x$ bijects

$$\{u \leq m_2 : |u| = d, u(x) \geq c\} \quad \text{with} \quad \{v \leq M_{red} : |v| = d - c\},$$

and if $d < c$ the first set is empty. Hence the degree sequence of Γ is

$$H(d) := h_\Gamma(d) = h_{m_1}(d) + \mathbf{1}_{d \geq c} h_{M_{red}}(d - c). \quad (2)$$

We want to prove that $H(d-1)H(d+1) \leq H(d)^2$ for $0 < d < e$.

For an integer-valued sequence $F : \mathbb{N} \rightarrow \mathbb{Z}$ set

$$\Delta F(0) = F(0), \quad \Delta F(d) = F(d) - F(d-1) \quad (d > 0),$$

$$\Delta^2 F(0) = \Delta F(0), \quad \Delta^2 F(d) = \Delta F(d) - \Delta F(d-1) \quad (d > 0).$$

For a single monomial $m = (a, b, c)$, let $A = a + 1$, $B = b + 1$, $C = c + 1$. A direct box count gives

$$h_m(d) = H_{A,B,C}(d) := \sum_{k=0}^d \mathbf{1}_{d-k < C} L_{A,B}(k),$$

where

$$L_{A,B}(k) = \#\{r < A : k - r < B, r \leq k\}.$$

The first-difference identities are

$$\Delta L_{A,B}(d) = \mathbf{1}_{d < A} + \mathbf{1}_{d < B} - \mathbf{1}_{d < A+B} \quad (d > 0),$$

$$\Delta H_{A,B,C}(d) = L_{A,B}(d) - \mathbf{1}_{d \geq C} L_{A,B}(d - C) \quad (d > 0).$$

Consequently, for $d > 0$,

$$\begin{aligned} \Delta^2 h_m(d) &= Z(A, B, C, d) \\ &= \mathbf{1}_{d < A} + \mathbf{1}_{d < B} + \mathbf{1}_{d < C} - \mathbf{1}_{d < A+B} - \mathbf{1}_{d < B+C} - \mathbf{1}_{d < C+A} + \mathbf{1}_{d < A+B+C}. \end{aligned} \quad (3)$$

In particular $Z(A, B, C, d) \leq 1$, hence $\Delta^2 h_m(d) \leq 1$ for every d ; the same is true for the shifted sequence $d \mapsto \mathbf{1}_{d \geq c} h_{M_{red}}(d - c)$. Thus

$$\Delta^2 H(d) \leq 2. \quad (4)$$

Moreover, if $\Delta^2 h_m(t) = 1$, then either

$$t \leq m(0), \quad t \leq m(1), \quad t \leq m(2), \quad (L)$$

or

$$t \geq m(0) + m(1) + 2, t \geq m(1) + m(2) + 2, t \geq m(0) + m(2) + 2, t \leq |m| + 2. \quad (U)$$

This is just (3) expanded with $A = m(0) + 1$, $B = m(1) + 1$, $C = m(2) + 1$.

We shall also use the following elementary consequences of the same box count. If $t \leq m(i)$ for all i , then

$$2h_m(t) = (t+1)(t+2),$$

and the same hypothesis (which also implies $t-1 \leq m(i)$, so the closed form is available at both t and $t-1$) gives

$$\Delta h_m(t) = t+1.$$

If $t \geq m(i) + m(j)$ for every pair $\{i, j\}$ and $t \leq |m| + 2$, then, with integer arithmetic,

$$2h_m(t) = (|m| + 1 - t)(|m| + 2 - t), \quad (5)$$

and if $t \geq m(i) + m(j) + 1$ for every pair and $t \leq |m| + 2$, then

$$\Delta h_m(t) = -(|m| + 2 - t). \quad (6)$$

Finally, for all t one has the universal upper estimate

$$2h_m(t) \leq (|m| - t + 1)(|m| - t + 2), \quad (7)$$

and if $2t \geq |m| + 1$, then $\Delta h_m(t) \leq 0$.

Now fix $0 < d < e$. Expanding $H(d) = H(d-1) + \Delta H(d)$ and $\Delta H(d+1) = \Delta H(d) + \Delta^2 H(d+1)$ gives

$$H(d)^2 - H(d-1)H(d+1) = (\Delta H(d))^2 - H(d-1)\Delta^2 H(d+1). \quad (8)$$

Since $H(d-1) \geq 0$ and $\Delta^2 H(d+1) \leq 2$, it is enough to prove

$$\Delta^2 H(d+1) = 1 \Rightarrow H(d-1) \leq (\Delta H(d))^2, \quad (9)$$

$$\Delta^2 H(d+1) = 2 \Rightarrow 2H(d-1) \leq (\Delta H(d))^2. \quad (10)$$

Indeed, if $\Delta^2 H(d+1) \leq 0$, (8) is immediate; if it is 1 or 2, (9) or (10) applies.

Write

$$f(t) = h_{m_1}(t), \quad g(t) = \mathbf{1}_{t \geq c} h_{M_{red}}(t - c),$$

so that $H = f + g$. Because $\Delta^2 f \leq 1$ and $\Delta^2 g \leq 1$, if $\Delta^2 H(d+1) = 1$, then either

$$\Delta^2 f(d+1) = 1, \Delta^2 g(d+1) = 0, \quad (A)$$

or

$$\Delta^2 f(d+1) = 0, \Delta^2 g(d+1) = 1. \quad (B)$$

If $\Delta^2 H(d+1) = 2$, then $\Delta^2 f(d+1) = \Delta^2 g(d+1) = 1$.

Case A, lower region for f . If $d+1 \leq m_1(i)$ for every i , then in particular $d+1 \leq m_1(x) < c$. Thus $g(d-1) = g(d) = 0$ and $\Delta g(d) = 0$. Hence $H(d-1) = f(d-1)$ and $\Delta H(d) = \Delta f(d)$. The lower formula gives

$$2f(d-1) = d(d+1), \quad \Delta f(d) = d+1.$$

Therefore $f(d-1) \leq (d+1)^2 = (\Delta f(d))^2$, proving (9).

Case A, upper region for f . Put $M = e = |m_1|$ and $U = M - d$. From the upper-region inequalities for f at $d+1$, $U \geq -1$, and (5)–(6) give

$$2f(d-1) = (U+2)(U+3), \quad \Delta f(d) = -(U+2).$$

Set $S = g(d-1)$, $S' = g(d)$, and $\delta = \Delta g(d) = S' - S$. The shifted summand satisfies

$$\delta \leq 0, \quad 2S \leq (U+2)(U+3), \quad 2S' \leq (U+1)(U+2), \quad (11)$$

which follows from (7) and the past-midpoint monotonicity applied to M_{red} after the shift. Indeed, if $d \geq c$, then the upper-region inequalities for f imply $2(d-c) \geq |M_{red}| + 1$, so midpoint monotonicity applies to the shifted M_{red} -sequence; if $d < c$, the shifted term is zero. It remains to show

$$f(d-1) + S \leq (-(U+2) + \delta)^2.$$

After multiplying by 2, this follows from

$$(U+2)(U+3) + 2S \leq 2(-(U+2) + \delta)^2. \quad (12)$$

Let $V = U + 2 \geq 1$. Since $S = S' - \delta$, the difference between the right and left sides of (12) is bounded below by

$$2\delta^2 - 4V\delta + 2\delta = 2\delta(\delta - 2V + 1) \geq 0,$$

using $\delta \leq 0$ and $V \geq 1$. Hence (9) holds in this case.

For the two cases in which $\Delta^2 g(d+1) = 1$ we use the following shift reduction. If $\Delta^2 H(d+1) = 1$ and $H(d+1) \leq (\Delta H(d+1))^2$, then

$$H(d-1) \leq (\Delta H(d))^2. \quad (13)$$

Indeed, $\Delta H(d) = \Delta H(d+1) - 1$ and $H(d-1) = H(d+1) - \Delta H(d+1) - \Delta H(d)$, so

$$H(d-1) \leq (\Delta H(d+1))^2 - 2\Delta H(d+1) + 1 = (\Delta H(d))^2.$$

Case B, lower region for g . Let $D = d + 1$. Since $\Delta^2 g(D) = 1$, necessarily $D \geq c$ and $\Delta^2 h_{M_{red}}(D - c) = 1$. In the lower subcase, $D - c \leq M_{red}(i)$ for every i . Set

$$X = D - c + 1 \geq 1.$$

Then

$$2g(D) = X(X+1), \quad \Delta g(D) = X. \quad (14)$$

Because $\Delta^2 f(D) = 0$, equivalently $Z(m_1(0) + 1, m_1(1) + 1, m_1(2) + 1, D) = 0$. Together with the lower inequalities for M_{red} , this implies

$$D < m_1(0) + m_1(1) + 2, \quad D < m_1(1) + m_1(2) + 2, \quad D < m_1(0) + m_1(2) + 2. \quad (15)$$

Since $Z(D) = 0$ and (15) makes all three pair indicators and the triple indicator equal to 1 at D , formula (3) shows that exactly two of the three single indicators are equal to 1 at D . For every $1 \leq k \leq D$, those same two single indicators remain equal to 1, while the three pair indicators and the triple indicator are still equal to 1. Hence

$$Z(k) \geq 2 - 3 + 1 = 0$$

Therefore $\Delta^2 f(k) = Z(k) \geq 0$ for $1 \leq k \leq D$.

The following two bounds on f at D then follow:

$$f(D) \leq \Delta f(D)(D+1), \quad \Delta f(D) \geq c. \quad (16)$$

Indeed, the first is the mean-value bound, valid whenever $\Delta^2 f \geq 0$ on $\{1, \dots, D\}$ (proof by induction on D : the inductive step uses $\Delta f(k-1) \leq \Delta f(k)$, then $f(k) = f(k-1) + \Delta f(k) \leq k \Delta f(k-1) + \Delta f(k) \leq (k+1) \Delta f(k)$).

For the second bound, permute the variables so that x is the third coordinate. Write

$$A = m_1(y)+1, \quad B = m_1(z)+1, \quad C = m_1(x)+1 = c, \quad \{x, y, z\} = \{0, 1, 2\}.$$

The first-difference formula for a principal sequence gives

$$\Delta f(D) = L_{A,B}(D) - L_{A,B}(D-C),$$

since $D \geq C$. From $Z(D) = 0$, the three pair indicators equal 1, and $D \geq C$, the two remaining single indicators equal 1. Thus

$$D < A, \quad D < B.$$

Consequently

$$L_{A,B}(D) = D+1, \quad L_{A,B}(D-C) = D-C+1.$$

It follows that

$$\Delta f(D) = (D+1) - (D-C+1) = C = c.$$

In particular,

$$\Delta f(D) \geq c.$$

Let $S_1 = \Delta f(D)$. Since $D+1 = X+c$, (14)–(16) give

$$2H(D) \leq 2S_1(D+1) + X(X+1) = 2S_1(X+c) + X(X+1).$$

The elementary inequality

$$2S_1(X+c) + X(X+1) \leq 2(S_1+X)^2$$

holds because its right minus left side is $2S_1(S_1-c) + 2S_1X + X(X-1) \geq 0$. Thus $H(D) \leq (\Delta H(D))^2$, and (13) proves (9).

Case B, upper region for g . Again let $D = d + 1$ and put $U = M - D$. The upper-region inequalities for $h_{M_{red}}$ at $D - c$ give $U \geq -2$, and (5)–(6), using $|M_{red}| = M - c$, give

$$2g(D) = (U + 1)(U + 2), \quad \Delta g(D) = -(U + 2). \quad (17)$$

For the first summand we do not need to know that D is in the upper region of f . The universal upper bound for a principal sequence gives

$$2f(D) \leq (M - D + 1)(M - D + 2) = (U + 1)(U + 2).$$

It remains to justify the sign of $\Delta f(D)$. Write

$$R = |M_{red}| = M - c, \quad q = D - c.$$

Since g is in the upper region at D , the upper-region inequalities for M_{red} give

$$q \geq M_{red}(0) + M_{red}(1) + 2, \quad q \geq M_{red}(1) + M_{red}(2) + 2, \quad q \geq M_{red}(0) + M_{red}(2) + 2.$$

Adding these three inequalities yields

$$3q \geq 2R + 6,$$

and hence $2q \geq R + 1$. Therefore

$$2D = 2(q + c) \geq R + 1 + 2c = (M - c) + 1 + 2c = M + c + 1 \geq M + 1.$$

Thus D is at or past the midpoint of the principal sequence $f = h_{m_1}$, so midpoint monotonicity gives

$$\Delta f(D) \leq 0.$$

Consequently

$$2f(D) \leq (U + 1)(U + 2), \quad \Delta f(D) \leq 0. \quad (18)$$

Set $S_1 = -\Delta f(D) \geq 0$. From (17)–(18),

$$2H(D) \leq 2(U + 1)(U + 2), \quad \Delta H(D) = -S_1 - (U + 2).$$

Since $U \geq -2$ and $S_1 \geq 0$,

$$(U + 1)(U + 2) \leq (S_1 + U + 2)^2$$

(the cases $U = -2, -1$ give LHS = 0, and for $U \geq 0$ we have $(U + 1)(U + 2) \leq (U + 2)^2 \leq (S_1 + U + 2)^2$ since $U + 1 \leq U + 2$, $U + 2 \geq 0$, and $S_1 \geq 0$). Hence $H(D) \leq (\Delta H(D))^2$, and (13) proves (9).

The case $\Delta^2 H(d + 1) = 2$. Then $\Delta^2 f(d + 1) = \Delta^2 g(d + 1) = 1$. Since $\Delta^2 g(d + 1) = 1$, we have $d + 1 \geq c$. Therefore the lower region for f is impossible, because it would give $d + 1 \leq m_1(x) < c$. So f is in its upper region. The lower

region for g is also impossible: if $d + 1 - c \leq M_{red}(y)$ for all y , then choosing any $y \neq x$ gives

$$d + 1 \leq m_1(x) + 1 + m_2(y) \leq m_1(x) + 1 + m_1(y),$$

whereas the upper-region inequalities for f give $d + 1 \geq m_1(x) + m_1(y) + 2$. Thus g is also in its upper region.

Let

$$U_1 = M - d, \quad U_2 = |M_{red}| - (d - c).$$

Note that $U_1 = U_2$, since $|M_{red}| = M - c$ gives $U_2 = M - c - (d - c) = M - d = U_1$; we keep two names for clarity in the algebraic identity below, which is true for arbitrary $U_1, U_2 \geq -1$. The upper-region bounds for f and g give $U_1 \geq -1$ and $U_2 \geq -1$ respectively. Applying (5)–(6) to f and to the shifted g gives

$$2f(d - 1) = (U_1 + 2)(U_1 + 3), \quad \Delta f(d) = -(U_1 + 2),$$

$$2g(d - 1) = (U_2 + 2)(U_2 + 3), \quad \Delta g(d) = -(U_2 + 2).$$

Therefore (10) follows from

$$(U_1 + 2)(U_1 + 3) + (U_2 + 2)(U_2 + 3) \leq ((U_1 + 2) + (U_2 + 2))^2.$$

The difference between the right and left sides is

$$2(U_1 + 1)(U_2 + 1) + U_1 + U_2 + 2 \geq 0,$$

because $U_1, U_2 \geq -1$.

Combining the cases proves (9) and (10), hence by (8) the sequence H is log-concave at degree d . Since $0 < d < e$ was arbitrary and H is the pure O -sequence of Γ by (2), the theorem follows.

References

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